

n-Gram based Language Separation for Minority Languages

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1 Introduction

Text data in minority languages are often limited in volume and on top of that mixed with snippets of majority languages because the source documents include articles or advertisements for the general population (Bartels 2020). This can result in minority text data that is too muddy to be analysed broadly. Every available text is valuable, but a lot of added text is unrelated information. In the long run, this makes it difficult to document minority languages using dictionaries or text corpora which in turn makes it difficult to implement reliable automated language analysis tools. This can lead to real life consequences that can hinder the revitalization of minority languages.

We can consider a simple analytical tool that has obvious real-life benefits for language users: a spelling checker. Spelling checks work by comparing an input to a predefined set of words (*bag of words*). The training data must be reliable in determining if an input is acceptable for a language while being efficient enough to be used fluently, i.e. it should not include more sophisticated considerations like word contexts. The word *Medizin* translates to *medicina* (Scientific Field) or *gójaŕstwo, lěkaŕstwo* and *medicina* (Medication) in Lower Sorbian.¹ The word *Medizin* is not actually Lower Sorbian and should be marked as a typo of *medicina*. Yet the corpus search returns entries for *Medizin*², indicating that it is considered a Lower Sorbian word.

¹ <https://dolnoserbski.de/dnw/woerterbuch/medizin>

² <https://dolnoserbski.de/korpus/komfort/pytanje/?wuraz=medizin>

Napšašowanje »medizin«, wuslědki 1–5 wót 5.

1	K tomu groni medicinaf Herwig Kollaritsch: „In der Medizin kann man weniges mit Sicherheit ausschließen, aber das schon: Eine Inkorporation der mRNA in unsere DNA ist technisch unmöglich“ (Žrědło:	DSB-NOW-2021 Nowy Casnik 72 (2021) 09, b. 2
2	Won grajo sobu w mustwje, kenž sejžecy balo grajo (Sitzball) – pši BSG " Medizin " w Chošebuzu –, tak podobnje ako pši pšćowem balu abo volleybalu.	DSB-CAS Nowy Casnik 37 (1985) 48, b. 4
3	W Serbskem kulturnem centrumje Slěpe pokažu waltoru, 31. julija, zeger 9.00 małym žišam film „Der Maulwurf und die Medizin “ a „Die Geschichte vom Fischer und seiner Frau“.	DSB-CAS Nowy Casnik 64 (2012) 30, b. 7
4	<< >> wot 7-11 a wot 3-5 we tom nowem Simonowem domje Brunšchwigu na goře. W Chošchebufu, 9. April 1870. Hermann Rath, Dc. der Medizin u. Chirurgie, geprüfter homöopathischer Arzt. Žiŕčeze wofyki k nahuberaŕu (wotemŕto gotowane) fa te tuŕnfche plaŕcheŕa, koŕche k drogowaŕu a	DSB-HIST bramborski-serski-zassnik-1870-18 
5	<< >> Plowkowy Ana Nowakojz Wilhelmina Mehliŕchojz Ana Lederojz Marjana Smogeriz Ana Lehmanojz. Rězniza. Nowe ŕłowa: Złoweznofč, Humanitāt, Menŕchlichkeit; huŕtrowidło, Medizin , Medikament, Heilmittel, teke: huŕtrowiza; kupadło, das Bad, das man nimmt; kupadliŕhczo, das Bad, in das	DSB-HIST wossadnik-sa-dolno-luzyske-wossady-1912-08 

Figure 1: The result set of the Lower Sorbian corpus search for the word *Medizin*.

Additionally, the text corpus contains at least 30 documents that contain the word *Medizin* but are listed as *dsb* language in the metadata³, even though the word *Medizin* is only present in German text snippets. This means that a spelling check that is implemented using a wordlist based on the text corpus from the Sorbian Institute would not be able to identify *Medizin* as a typo of the actual Lower Sorbian word *medicina* because it will use the data from the text corpus as training data. In consequence, an automatic spell check cannot reliably tell whether *Medizin* should be considered correct or a typo. This translates to a missing clarity for language users which in turn potentially solidifies this problem further.

Since more sophisticated automatic methods for text mining that may include POS tagging, lemmatization, context analysis, word relations and generally more complex input rely on clean training data to produce clear results, noisy data gates a lot of research potential from being applied to minority languages.

Therefore, research institutes like the Sorbian Institute are putting major effort into separating the minority language data – Lower or Upper Sorbian in this case – from the majority language text snippets – in this case German. This is often done using manual work and text annotation (Sorbisches Institut 2022). Generic methods that help researchers to automatically tidy up the data by marking the majority language elements will surely help with that task.

This work describes a trigram-based (Biemann, Heyer, & Quasthoff 2022) approach for majority language detection and combines it with a document structure URN notation (Tiepmar & Heyer 2019) to identify the structural text elements that are supposedly not part of the minority language. The goal is to be able to clean the text data for broad analysis and provide candidate lists for text passages that

³ *dsb*, *deu* exists in the metadata to signal mixed language documents.

might be considered for deletion or to enrich the language metadata annotation. The approach is designed to be – aside from the ability to create n-Grams - independent from the minority language to make it applicable for other minority languages as well. The method is also designed to only use basic text mining and programming skills in a self-sufficient manner; there is no sophisticated software or hardware required to repeat this algorithm in other scenarios. The complex issue of spelling variation is considered in the algorithm description.

2 Algorithm

An algorithm that achieves the described goal can generally be approached in three ways but because of the data sparseness of minority languages, only one is realistic. Detecting the minority language or deciding a result based on probability cannot really be done as it would require clean minority language training data. This limits the approach the third option of detecting the majority language. To avoid unnecessary data loss, this approach should be designed in a way that focuses on high precision because there is no way to counterbalance the generation of false positives as it would be possible if configurable probability thresholds were generated. This is also the reason why the results should not be considered as final but merely as candidates that need to be checked manually.

The method for language separation that is implemented in this study can be divided into 3 individual steps. At first the data and training data must be formatted correctly. Then a list of trigger words for the majority language are generated based on trigrams and in the final step a list of text passages is retrieved that include these trigger words. This list of references for text passages is the desired result of the algorithm.

2.1 Step 1: Preparation

At first it is necessary to have a considerable amount of majority language words - in this case German. This study uses 4 wordlists that aggregate to a combined list of 1'943'567 tokens based on the Textgrid Archive (TextGrid Konsortium 2006–2014), the German Political Speeches Corpus (Barbaresi 2019), The ASV Wortschatz wordlist (Quasthoff & Richter 2005) and the wordlist of the IDS Mannheim (IDS 2025). Tokens with length 1 were ignored. To cover spelling variations, every word was additionally added with the characters *f* and *f* replaced by *s*. Lower Sorbian writing includes much more spelling variations but since normalization is added to the text corpus annotation – and therefore available for future work - and each included generated variation increases the fuzziness of the evaluation results, only this example was implemented as an indicator for a possible practical solution.

Token frequencies are not required.⁴

The minority language texts must be converted into a list of n-Grams. Trigrams⁵ were chosen for this work, but any type of n-Gram should be applicable as well. Higher n increases the weight of the individual words contexts which should result in more limited but also more precise results. It is worth considering that a generated n-Gram should generally not exceed sentence punctuation which means that higher n imply that a lot of text snippets will be ignored as the surrounding sentence or subordinate clause is simply too small to produce an n-Gram.

n-Gram frequencies are not required for the basic method but may be useful for evaluation and future improvements.

With a majority language wordlist and a minority language list of n-Grams, the preparations are complete, and the core of the algorithm can be implemented.

2.2 Step 2: Language Scoring

The goal of this step is to produce a list of trigger words that have a high – or even definite – probability of indicating majority language text snippets. These trigger words will then be used as input in the final step.

With a majority language wordlist *tokens_maj* and a minority language list of n-Grams *ng_min*, it is possible to determine, what portion of each individual n-Gram *ng_i* is counted as majority language. If this portion is 100% - meaning that each token in *ng_i* is included in the majority language wordlist - then *ng_i* has a high likelihood of being an n-Gram of the majority language. The result is an integer language score ranging from 0 to n for each n-Gram.

```
ng_n = {}
ng_lower = {}
foreach ng_i in ng_min:
    . score = 0
    . foreach token in ng_i:
        . . if token in tokens_maj:
            . . . score = score + 1
    . if score == n:
        . . ng_n [ng_i] = score
    . else:
        . . ng_lower [ng_i] = score
```

⁴ Token frequencies may be useful for more advanced methods. For instance they allow the inclusion of weighting or make it possible to ignore tokens that are very frequent in the minority language but rare in the majority language but somehow still skew the result.

⁵ n-Grams with n=3

If the minority language is characterized by heterogenous spelling variations, the tokens should be normalized in the same way then the generated variations in the wordlists when compared. Spelling variations are generally a complex issue, and it is difficult to estimate best practises but, in this study, normalizing the tokens for comparison and keeping the n-Grams as they appear in the document worked well. Changing the n-Grams may lead to inconsistencies in the third step because it may produce trigger words that do not appear in the documents.

n-Grams with language score of n indicate majority language tokens that are used in majority language contexts but may still also be used in minority language contexts indicated by their presence in n-Grams with a lower language score. This means that the desired list of trigger words can be created as a list of tokens that appear in n-Grams with language score n but never in n-Grams with a lower score.

```
trigger = {}  
foreach token in ng_n:  
  . if not token in ng_lower:  
    . . trigger[token] = 1
```

The resulting wordlist *trigger* is composed of majority language tokens that appear exclusively in majority language contexts and therefore are very likely to indicate majority language text snippets based on the knowledge that is the majority language wordlist. With this list of trigger words it is trivial to implement the last step of this method.

2.3 Step 3: Text Snippet Retrieval

The list of trigger words that was generated in the previous step can be used to mark text passages that are likely to include majority language text snippets by searching for token positions of each individual trigger word. The result can be generated as a list of context snippets or sentences or structural elements depending on what is more applicable. To limit unnecessary data loss, it is reasonable to produce the resulting list as fine-grained as possible and not include too much surrounding text context around the trigger words but generally the output depends on what is desired and possible in specific setups. Since the Lower Sorbian text corpus is published as part of the Canonical Text Service Infrastructure project (Tiepmar, 2025) and CTS URNs themselves provide references for fine-grained text snippets that fit well into such workflows, the result of this step is a list of CTS URNs that is generated by searching for the trigger words in all documents using the publicly available *structurebagofwords* function that provides token list representations of the individual structural text elements.

3 Evaluation

To evaluate this approach, the complete Lower Sorbian text corpus - including the copyright restricted documents - was analysed using the progress of February 2025. Future updates will influence the list of trigger words and shift the results a little, but the effect should not be strong enough to invalidate this evaluation. Because of copyright restrictions, the text corpus cannot be made publicly available. Instead, the identified text passages are added as token lists.

3.1 Precision

Each text passage that was identified in Step 3 was manually checked. Most passages proved to be clearcut decisions, but it is worth considering that some cases are not as trivial. This can include roman numbers or named entities. Specialized job labels and titles like *feldmarschall* can be especially tricky to evaluate, since they can be argued as German words but also as the job label that is used in Lower Sorbian. These unclear cases were few and the results are meant to be manually checked anyway, which mitigates this problem. Nevertheless it is worth pointing out that the decision, whether an individual text passage was correctly identified as German, includes some bias.

3.2 Recall

Since the texts are not fully annotated and the annotation may be incomplete, recall cannot be calculated reliably. Yet it is possible to calculate the coverage of already manually identified text passages based on the available text metadata. This was done by comparing the list of URNs from Step 3 to the list of URNs that are labelled as *de*, *deu* and *lat* in the text corpus.

3.3 Generic Evaluation

The generic method identified 25'670 text passages in 3'371 documents. The 25'670 text passages were manually checked and 846 false positives found, resulting in a precision of 96,7%, a value that is quite respectable considering that it was achieved without any minority language knowledge and with basic text mining techniques. This implies that similar results can be achieved in other language configurations as long as a comprehensive list of majority language words is available.

When comparing the resulting text passages with the text passages that were already marked in the text metadata, it showed that 8'082 of them were new candidates and 17'588 already either marked directly or indirectly through the annotation of a parent structure element. The fact that a lot of the text passages were already manually marked indicates a certain reliability of the algorithm.

32'820 trigger words were generated that can be reused as corpus specific indicators for German text in Lower Sorbian documents. Based on the false positive text passages, a list of 532 wrong trigger words were generated, that will be reused in the following description of practical tweaks of the

algorithm. Whether these trigger words should be reused and supplemented with trigger words of future iterations is hard to say but since they are generated based on a very strict method, reusing them appears reasonable even if they do not appear in future iterations of this workflow.

The introductory example *Medizin* is unfortunately still not recognised as German because of one n-Gram *der-medizin-u*, which appears one time in the full text corpus and is a German n-Gram⁶. But since tokens with word length 1 were not added as majority language training data, the algorithm does not recognise the token *u* and therefore an n-Gram with language score n-1 exists for the word *medizin*.

4 Tweaking

This chapter explains, how the input can be tweaked based on the intermediate results to improve the result in two ways. Since both improvements are based on personally handpicked changes, the evaluation is no longer scientifically valid but can still provide insight into improvement potential in practical usage. The tweaking can happen in two ways represented by the following two subchapters: The number of false positives can be reduced to increase precision, and the number of majority language words can be raised to increase recall. Both changes can then be applied incrementally until no further improvement is visible.

Both tweaks can technically be done with just the intermediate results and knowledge about the majority language but benefit strongly from expert knowledge about the minority language.

4.1 Adding False Friends

Based on the list of false positives, a list of wrongly suggested trigger words – labelled as false friends in the following description - can be generated. These false friends are then ignored when loading the majority language word list resulting in a targeted reduction based on the intermediate result. This list can be supplemented with words that are known as false friends if language knowledge is available. The word *nicht* is a good example since it looks atypical for Lower Sorbian but exists in the Lower Sorbian language⁷ and therefore should not be allowed as a trigger word. Whether individual entries should be considered false friends is a biased decision and sometimes – for instance *gefchicht* – questionable but it should not make a significant difference in the context of the description of this method.

⁶ Text passage: ... der Medizin u. Chirurgie, geprüfter homöopathischer Arzt ...

⁷ See <https://dolnoserbški.de/ndw/pokaz/Muka/nicht/nicht/nicht> the sentence *ja som južor słyšaŭ, aŭ hu was nicht doma nějo* which translates to German *ich habe bereits gehört (erfahren), dass bei euch niemand zu Hause ist*, with the word *nicht* translating to *niemand*.

In this study 532 false friends were generated based on the false positive results reducing the length of the majority language input wordlist by 530 from 1'943'567 to 1'943'037. The gap between 530 and 532 can be explained by the beforementioned variations of the characters *ſ*, *f* and *s*, that are dynamically added while the wordlist is generated. This means that some false friends do not appear in the initial wordlist while others reduce the wordlist by more than one, making the correlation between the wordlist reduction and the number of false friends a little fuzzy. 82 false friends appeared in this study that are not included in the initial wordlist, each being an example case of the implemented normalization⁸.

Since no new trigger words are generated, no new text passages get identified. The false positives from before cannot reappear because the trigger words for these were used to generate the list of false friends. In consequence, none of the registered and no new false positives can appear in the result set, reducing the number of false positives to zero. The inclusion of the false friends reduced the result set from 25'670 to 24'571 text passages. Interestingly, the reduction by 1'099 is higher than the number of false positives from the generic analysis. The number of documents fell from 3'371 to 3'065 documents. As expected, no new text passages were identified and none of the results is interpreted as a false positive, resulting in a pseudo-precision of 100%. This should not be misinterpreted as a precision of 100% in a scientific sense because the result was generated by manually excluding every item that was misclassified in the generic approach. Anything other than a precision of 100% at that point would be an unexpected phenomenon.

The number of new text passages compared to the language metadata annotation reduced from 8'032 to 7'166 while the number of already known text passages fell from 17'588 to 17'405.

32'111 trigger words were generated, a reduction of 709 from the 32'820 previous trigger words. This reduction is higher than the number of false friends, which is to be expected because the false friends impact the number of relevant n-Grams. The difference of 177 between the reduction of trigger words and the number of false friends indicates that the collateral damage from including the false friends is negligible when compared to the benefit of the reduced error rate. No new trigger words were generated, and all generated trigger words were already known.

Generally, the evaluation of the inclusion of the list of false friends shows exactly the results that were expected. At this point, no new trigger words or text passages should appear in that result set

⁸ adminiftrator, befeke, caftner, efki, fefch, fonfeca, froefe, gefchicht, graf, gutfch, hradfchin, jörgenfen, jufab, kiekebufch, konfervator, kofelleck, kofta, kreif, lefem, loeft, löfer, meifner, mouftier, neyf, nicholfon, nifu, paulowfky, petfchaft, petfchek, pflugfchar, piftol, plüfkow, poffeffivum, proceffu, prüf, refeda, rift, rundftedt, sabifch, schnurgaffe, foftatus, suf, sufuf, swanfea, syftem, fabio, famet, fchawe, fchee, fchpil, fcht, fenn, fermone, fhech, filbr, folitär, fozialismus, fpeifen, fpok, ftaatsfecretär, ftaby, ftela, fterl, ftruf, fultana, fupernumerar, fyk, texaf, thefaurus, tifcher, truchfeß, tfchernowitz, turfo, verfammlunga, vettera, vogefen, wahlftatt, webfter, werfche, weftftraße, witkowi, zeugfchmied

and the identified errors should disappear. Therefore, it is not surprising that the word *medizin* could not yet be identified.

4.2 Adding Found Majority Language Tokens

Recall can be improved by adding manually selected words that are not yet included in the majority language wordlists and therefore considered as minority language indicators by the algorithm. n-Grams with language score $n-1$ can provide useful additional candidates as well as expert knowledge or preexisting curated minority language wordlists. Tokens that appear only once in the text corpus can help finding atypical spellings and corpus specific typos, but it is not recommendable to generate such a list automatically just by token frequencies.

In this study, 531 words could be identified including rare words (*wollverkaufsgeschäft*), atypical spellings (*wirthschaft*, *schwarze*) and corpus specific typos (*umtauschgeschäftvon*, *mikteilungen*). The method of acquiring this list was a diffuse mixture of result screenings of iterative test runs and language expert feedback.

32'563 trigger words were generated, an increase of 452 from the 32'111 trigger words generated when false friends were included. This correlates perfectly with the difference to the generic method with 32'111 known and 452 new trigger words.

Newly generated trigger words are expected to produce new text passage candidates, and it is not surprising that the number of identified text passages lies between the previous two approaches with 25'170. Compared to the generic analysis 598 new text passages could be identified with 71 new documents and 599 new text passages with 84 new documents when compared to the inclusion of false friends. The number of documents is 3'149, a value between the 3'371 and 3'065 from the previous two results. The number of false positives sank from 846 to 20, resulting in a heavily biased pseudo-precision of 99.9%. None of the false positives was already marked in the language metadata annotation.

While being scientifically invalid and heavily biased to the point of being fraud, the resulting evaluative values between the results from the generic method and the inclusion of the false friends indicate that both tweaks work as intended and can be valuable in a practical setting. The inclusion of false positives reduces the error rate to zero without any recall benefit if done as described in this study. The inclusion of majority language words based on the intermediate results increases recall while staying sensitive for false positives if it is done while still respecting the false friends list.

With the false positives and the manually selected found German words including the single character word *u*, the word *Medizin* is now correctly identified as 100% German. However, when repeating this analysis with a more up to date version of the text corpus, it disappears again from the list of trigger words, hinting at two pitfalls of this algorithm that are worth considering.

5 Pitfalls

Two major pitfalls of this method might have been already spotted by the attentive reader in the dichotomy between the evaluation of the introductory example word *Medizin*, which was identified successfully, and the example image in Figure 1, which lists the example sentence *Won grajo sobu w mustwje, kenž sejžecy balo grajo (Sitzball) – pši BSG "Medizin" w Chošebuzu –, tak podobnje ako pši pšćowem balu abo volleybalu*. This example sentence should create trigrams for the word *Medizin* that are not fully German⁹, which would make it impossible for the word *Medizin* to become a trigger word. The fact that this word is still detected can be explained in two ways, each hinting at possible pitfalls.

Explanation one is that none of the *Nowy Cassnik* documents from 1985 were part of the text corpus when the data were collected for this analysis. This illustrates that the generated trigger words can change depending on different versioning states of the input data if the text corpus is not finalized. While this could be considered neglectable in practical scenarios because the trigger words can be expected to be reusable as language indicators, they should be recalculated if the result is expected to be consistent and additional documents are added. This even opens of the possibility to generate targeted trigger words based on predefined sub corpora.

The second explanation is that results can change drastically depending on the tokenization and type of generated n-Gram. In this study, n-Grams were generated only within text units fenced by punctuation¹⁰ and tokenization was implemented based on manually selected special characters¹¹ that were found in the data. Whether the word *Medizin* becomes a trigger word changes depending on whether " is considered punctuation that fences n-Grams. In the configuration used in this study, this is the case and therefore no conflicting n-Gram is generated because the token *Medizin* stands alone and does not form a suitable n-Gram. *Medizin* would stay a trigger word.

Finally, automated tweaking of the input and automatic text corpus alterations based on the output is not recommendable, especially with respect to the special characteristics of minority language data. There is a high risk of unnecessary data loss and the sparseness of training data makes it difficult to estimate the incorrectness of a given result entry. The results should only be considered as suggestions for candidates that need to be checked manually.

It can also be expected that the quality of the results depends on the similarity of the languages. In this study a slavic language was separated from a non-slavic language. It is reasonable to assume that

⁹ "Medizin" w Chošebuzu and pši BSG "Medizin" both contain tokens that are clearly not German language.

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the results would be worse if the task would have been to separate Lower Sorbian from – for instance – Polish text snippets. But this should be true for any algorithmic and non-algorithmic method of language classification – the more similar two languages are, the harder it is to differentiate them based on snippets.

6 Future Work

This method provides numerous ways to improve the results. Tweaking the wordlists for the majority language training data and false friends is a solid way to incrementally improve recall and precision based on previous results or fresh input. Depending on the data, it can also be considered that certain wordlists are more or less useful. For instance, the wordlist based on the Textgrid corpus might be more fitting for historical texts while the German Political Speeches corpus can be more helpful for modern texts. Normalization can be investigated by either repeating the process with tokens and n-Grams that use the annotated normalized spelling or automatically generated variations. Token and n-Gram frequencies might be useful for reducing the weight of exceptional cases like the singular n-Gram that prevented the token *Medizin* from being classified correctly. With the much-improved standardization of Lower Sorbian spelling, it will be interesting to see the reliability of this workflow when used on future iterations of the text corpus. The generated wordlists based on n-Grams with score 0 can be valuable input for language detection based on text mining and statistics. Finally, the identified text passages may help to improve the language annotation of the Lower Sorbian text corpus.

Scripts and evaluation results are published on Zenodo.¹² The scripts are specialized for this study and – especially since they rely on a specific setup – should not be considered as generically applicable but merely as documentation for the evaluation. Since the described analysis is using very basic text mining concepts, it is more recommendable to reimplement the algorithm for specific scenarios.

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Publication rejected for the following reasons:

Review 1

Originality of the paper:

From 1 (very poor) to 4 (excellent): 3

Comments: The aim of the paper looks interesting and innovative, but not always fully expressed

Formulation of a clear research question:

From 1 (very poor) to 4 (excellent): 2

Comments: The question can be somehow deducted in a however not well formulated introduction; part of the paper concerns a criticizable methodology

Familiarity with the research context and the relevant literature:

From 1 (very poor) to 4 (excellent): 3

Comments: While the mentioned literature is quite short, the familiarity with the research context is evident

Accuracy of the analysis and plausibility of the results:

From 1 (very poor) to 4 (excellent): 2

Comments: see below and notes

Examples and empirical support:

From 1 (very poor) to 4 (excellent): 2

Comments: see below and notes

Relevance and clarity of the title and the abstract:

From 1 (very poor) to 4 (excellent): 2

Comments: An abstract has not been provided

Overall, the topic explored in the article is of great interest to the field of corpus linguistics, and it is clear that the author has extensive knowledge of the subject. However, there are two methodological shortcomings, the combination of which prevents the article from being fully enjoyable. On the one hand, it is easy to deduce that the author, who is not a native English speaker, did not have the article copy-edited; some passages are rather obscure and difficult to understand (e.g. the opening of the second section) merely because of this non-fulfillment. On the other hand, in several passages the author takes the meaning of sectorial terminology or the content of entire paragraphs for granted, which may cause a general reader not to understand: when discussing sector-specific terms and processes, the author should not limit themselves to citing useful bibliography, but should provide some examples and/or brief explanations in footnotes, given that the volume is intended for an audience that is not always specialised in corpus linguistics. There are also various methodological and content-related fallacies, highlighted in the text of the article. The space devoted to chapters 4 and 5 (which the author himself identifies as dealing with results that are ‘scientifically invalid and heavily biased to the point of being fraudulent’, not the most pleasant statement in an academic study) is proportionally much greater than the one of the previous sections, which are generally rushed and lack impact, despite being the most relevant part of the paper. The use of Zenodo for sharing attachments should be avoided in future, as it clearly reveals the identity of the author of the paper during the review phase. It is not fully clear how these attachments should be used (a “readme” file should be enough), especially given the fact that the texts on which the analysis was conducted have not been made available for copyright reasons. In conclusion, the text provided by the author is overall closer to a draft rather than to an article and requires substantial and necessary amendments in terms of both language and content. I therefore advise the editors to accept the contribution – whose initial intentions are nonetheless valuable and really interesting – while giving the author due time to implement the aforementioned changes.

Review 2

Review of “n-Gram based Language Separation for Minority Languages”

The research presented in this article is potentially interesting, but unfortunately the text is very hard to follow. First, it lacks any introduction or empirical basis. No description of Sorbian is provided, either grammatically or sociolinguistically, while the individual chapters focus on empirical methods and data that are difficult to interpret. One wonders what the overall objective of the research is, what Sorbian’s situation is with respect to digital tools, and what the main contribution this study intends to make. The article should be completely rewritten to make it interesting and, above all, consistent with the overall profile of the volume. I wonder whether the author is willing to undertake such an in-depth revision. For this reason, I believe the article is unripe for publication.